

5.9.1. Network Recovery channels

Conceptually, the Network Recovery process is possible over any commissioning channel. As the Matter specification evolves, the designers SHOULD consider providing a Network Recovery feature for any new commissioning channel. In order to fully specify the Network Recovery procedure, commissioning channel definitions SHALL specify:

- an advertising format that distinguishes a network recovery advertisement from the commissioning advertisement. The advertisement for Network Recovery SHALL contain the value of the [RecoveryIdentifier](#) attribute and SHOULD contain the error codes that led to the need for network recovery.
- provisions for [Extended Announcement](#) of Network Recovery advertisements.
- an ability to establish a [CASE](#) session over the commissioning channel. That CASE session SHALL be used for the recovery of network connectivity and not for steady state operational interaction.

5.9.2. Implementer Considerations

When the network connectivity is lost — for example, resulting from equipment replacement, network reconfiguration, etc. — it is usually desirable to restore connectivity to all affected devices. In some cases, the loss of network connectivity is intentional with the user expecting not to reconnect some of the end devices or some of the controllers to the network. It MAY be impossible for an Administrator to discern the user's intent that led to the loss of connectivity, so Administrators SHALL seek user consent / intent to determine which Nodes would be subject to the Network Recovery process.

5.9.3. Network Recovery Flow

The Network Recovery process consists of the following steps enumerated below. Recovery Node refers specifically to the node that has lost its operational network connectivity. This process assumes that the operational network credentials for one or more Nodes has been changed such that those Nodes are no longer able to communicate with other Nodes, including Administrators. Normative requirements below apply to Nodes that choose to implement the Network Recovery feature.

1. At some time before the operational network credentials change, an Administrator SHOULD retrieve the value of the [RecoveryIdentifier](#) attribute from the GeneralCommissioning cluster on the Recovery Node. This MAY be done during the commissioning process.
2. The Administrator SHALL have the new credentials required to connect to the operational network, and the new operational network SHALL be present and operational.
3. Should operational network connectivity be lost, the Recovery Node SHALL continue to attempt to connect to the operational network for a duration of at least 120 seconds using its existing credentials. Afterwards it MAY choose to enter Network Recovery mode and commence announcement. Recovery Nodes SHALL attempt to reconnect to the existing operational network while in the recovery state. Given that temporary network outages occur with some frequency, waiting before entering Network Recovery mode and commencing announcement limits wireless spectrum pollution/interference and helps to limit attack opportunities.